胡琳

手机号码：14706824492 | **邮箱：1436254290@163.com** | **MBTI:**ISFP | **政治面貌：**预备党员

教育背景

华南农业大学（**双一流**） 电子科学与技术 **2023.9 – 至今**

- **专业成绩：**4.42/5 （6/100）
- **英语水平：**CET-4（529）、CET-6（487）
- **主修课程：**高等数学（99）、大学物理（98）、模拟电子技术（94.2）、C语言程序设计（94.5）、电路（93）
- **专业技能：**熟练掌握C、Verilog、Python等语言，精通**Keil、Vivado、ModelSim、Altium Designer、VScode、MATLAB**等软件
- **校园经历：校一等奖学金（2/514）、校优秀共青团员**、创芯实验室（ICIL）副会长、无人机协会干事

项目经历

基于YOLOv8的智能柚园巡检机器人 2024.5-2025.5

作为**团队负责人**，我统筹项目推进与团队管理。技术层面，我深度参与机械结构的设计，通过调研与仿真，提升设备对果园任务的适配性；另一方面，开发基于stm32的小车控制系统，完成**基础运动控制算法编写并实现精准喷药功能的逻辑闭环**。同时，我独立负责 LoRa 无线通信模块的集成与调试，解决了远距离数据传输中的信号干扰问题，实现小车作业数据与上位机的稳定交互。此外，为进一步提升项目智能化水平，我与成员在项目后期引入了 **YOLOv8 病虫害识别模型**，完成模型训练与轻量化部署，实现对柚子树苗生长状态的识别，同时还参与了交互式网页的设计。

项目最终成功获批**大学生创新训练项目立项**并以 “优秀” 等级提前结项；核心技术成果获 “柚园生长数据采集与分析平台”、“柚园水肥一体化智能灌溉系统”**两项软件著作权登记**，机械结构设计获 “一种智慧农业用柚子树苗灌溉装置” **实用新型专利授权。**

基于FPGA的多传感器融合天气图像处理系统 2025.5-2025.9

作为**图像处理算法负责人**，我设计并在 Zynq Artix-7 上完成了**端到端的高性能图像信号处理（ISP）方案**落地 —— 从 OV5640 图像传感器的数据采集，到最终 HDMI 显示输出的全链路流程中，开发并实现了噪声抑制、坏点校正、自动白平衡等 **22 种图像处理增强算法**，通过硬件层面的优化适配，显著提升了输出图像的清晰度、色彩还原度与整体视觉质量。在数据处理方面，我负责多源环境传感器数据处理，通过编写多字节收发串口程序，为系统搭建数据获取通道。**独立完成基于 PyQt6 的交互式上位机界面开发**，不仅支持对 FPGA 端图像算法的灵活切换，还实现了传感器数据可视展示。同时，与队友合作，在Zynq 7020上部署**基于MobileNet的天气图像识别模型。**

项目最终成功获全国大学生集成电路创新创业大赛二等奖，目前在投《集成电路与嵌入式系统》。

面向荔枝病虫害诊断与防治的多智能体协同系统 2025.5-至今

针对荔枝病虫害防治的自动化与智能化需求，我参与构建了面向荔枝病虫害诊断与防治的多智能体协同系统。算法优化方面，以**CLIP 预训练模型**为基础实现高精度病虫害识别，同时创新性引入**Faster R-CNN 网络**，有效解决了病虫害定位精度不足的关键问题；架构设计层面，独创性地以 **DeepSeek** 大模型为基座，设计多模态模块化定制框架，通过自研任务复杂度分类器，显著提升大模型在病虫害诊断与预防系统中的检索效率；整体方案为荔枝病虫害精准防治提供了核心技术支撑，直接推动果园管理效率提升与成本优化。

项目最终成功获**国家级大学生创新训练项目立项**，发表论文《基于YOLOv11n-ACE的轻量化荔枝FPN改进病虫害检测方法》，目前已被**农业机械学报录用。**

100%可复用火箭点火器 2025.4-2025.9

在北京冕巢航天科技有限公司实习五个月，作为嵌入式实习生，深度参与**火箭点火器控制系统的全流程研发**，独立负责**点火系统的时序逻辑设计与控制程序开发**，精准定义点火触发、状态监测、应急中断等关键时序节点，为系统稳定运行奠定基础。全程参与多轮点火试验，助力团队成功完成点火器产品研发。**独立开发点火控制系统上位机软件**，实现点火参数配置、实时状态监控、历史数据回溯等功能，该软件已成功获取计算机软件著作权；同时设计基于**九轴传感器与舵机的火箭姿态控制算法**，为火箭姿态稳定控制提供了技术参考。

荣誉奖项

第八届全国大学生嵌入式芯片与系统设计大赛FPGA创新设计赛道	国家级一等奖
第八届全国大学生嵌入式芯片与系统设计大赛芯片设计赛道	国家级二等奖
2025年中国大学生机械工程创新创意大赛	国家级三等奖
第十八届全国大学生节能减排社会实践与科技竞赛	国家级三等奖
第九届全国大学生集成电路创新创业大赛	华南赛区二等奖

…其他省部级奖项4项

Lynn Hoo

Phone number:	E-mail:	MBTI:	Guangzhou
14706824492	1436254290@163.com	ISFP	China

EDUCATION

South China Agricultural University (SCAU) **2023.9 – Present**

Bachelor of Engineering in Electronic Science and Technology

- **Academic Performance:** GPA: **4.42/5.0** (Rank: **6/100**)
- **Core Courses:** Advanced Mathematics (99), University Physics (98), Analog Electronics (94.2), C Programming (94.5), Electric Circuits (93).
- **English Proficiency:** CET-4 (529), CET-6 (487).
- **Honors & Awards:** University First-Class Scholarship (Twice, Rank: 2/514).Vice President of Integrated Circuit Innovation Lab (ICIL).Member of UAV Association

TECHNICAL SKILLS

- **Programming:** C, Verilog, Python.
- **Tools:** Vivado, ModelSim, Keil, Altium Designer, MATLAB, VS Code.
- **Hardware:** FPGA , STM32, RISC-V, Sensors & Actuators.

PROJECT EXPERIENCE

Lightweight Pest Detection Method for Lychee Orchards based on YOLOv11n-ACE

2025.04 – 2025.09

- **Core Researcher:** Developed an optimized pest detection model to improve precision in complex orchard environments.
- **Innovation:** Proposed the **YOLOv11n-ACE** architecture, incorporating lightweight FPN (Feature Pyramid Network) improvements to balance detection accuracy and computational efficiency for edge devices.
- **Outcome:** Published as the first author in the **Journal of Agricultural Machinery** (Accepted); Project officially funded as a **National Undergraduate Innovation and Entrepreneurship Training Program**.

Multi-Sensor Fusion Weather Image Processing System based on FPGA

2025.05 – 2025.09

•**Algorithm Lead:** Designed and implemented an end-to-end high-performance Image Signal Processing (ISP) solution on **Xilinx Zynq Artix-7**.

•**Hardware Optimization:** Developed 22 image enhancement algorithms including noise suppression, bad pixel correction, and automatic white balance to improve visual quality from OV5640 sensors to HDMI output.

•**System Integration:** Deployed a **MobileNet** weather recognition model on Zynq 7020 and developed an interactive host computer interface using **PyQt6** for real-time algorithm switching and data visualization.

•**Awards:** Won the **National Second Prize** in the National University Integrated Circuit Innovation and Entrepreneurship Competition; Paper currently under review at *Integrated Circuits and Embedded Systems*.

100% Reusable Rocket Igniter Control System

2024.12 – 2025.05

- **Embedded Systems Intern** at Beijing Mianchao Aerospace Technology Co., Ltd.
- **Design & Development:** Independently designed the timing logic and control programs for the ignition system, covering ignition triggering, status monitoring, and emergency interruption.
- **Software Development:** Developed the PC-side control software (Python-based) for parameter configuration and real-time monitoring; obtained a **Computer Software Copyright**.
- **Navigation & Control:** Designed a rocket attitude control algorithm utilizing 9-axis IMU sensors and servos.

Intelligent Pomelo Orchard Inspection Robot based on YOLOv8

2024.05 – 2025.05

- **Team Leader:** Managed overall project progress and interdisciplinary coordination.
- **Hardware & Control:** Designed the STM32-based motion control system and integrated LoRa wireless modules for long-distance, anti-interference data transmission between the robot and the host station.

HONORS & RESEARCH AWARDS

National Level

- **1st Prize:** 8th National University Embedded Chip and System Design Competition - FPGA Innovation Track.
- **2nd Prize:** 8th National University Embedded Chip and System Design Competition - Chip Design Track.
- **3rd Prize:** 2025 China University Mechanical Engineering Innovation and Creativity Competition.
- **3rd Prize:** 18th National University Student Social Practice and Science & Technology Contest on Energy Saving and Emission Reduction.

Provincial & Regional Level

- **2nd Prize:** 9th National University Integrated Circuit (IC) Innovation and Entrepreneurship Competition - South China Region (also known as the "Huanxin Cup").
- **2nd Prize:** 11th "Datang Cup" National University Mobile Communication Technology and 5G Application Innovation Competition - Guangdong Province.
- **2nd Prize:** 2025 Guangdong Graduate Academic Forum - "Automobile and Electronic Environmental Protection" Innovation Competition.
- **3rd Prize:** 17th "China Hardware Cup" (AIC) National University Computer Promotion and Application Selection Contest - Guangdong Province.
- **3rd Prize:** 10th "Challenge Cup" Guangdong University Student Science and Technology Academic Works Competition.